# Type MS · Explain stereoscopic vision and how it helps judge distance

## Answer 1 of 3 · Question 1 part b · 3 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 1 | (a) | <table><tr><th>Description</th><th>Part</th></tr><tr><td>contains light sensitive cells</td><td>retina;</td></tr><tr><td>vibrates when receives sound waves</td><td>ear drum;</td></tr><tr><td>helps the body balance</td><td>semi-circular canals;</td></tr><tr><td>controls the amount of light entering the eye</td><td>pupil;</td></tr><tr><td>transfers vibrations across the middle ear</td><td>ossicles;</td></tr><tr><td>turns vibrations into nerve impulses</td><td>cochlea;</td></tr><tr><td>refracts light entering the eye</td><td>cornea;</td></tr></table> | | 7 |
| | (b) | • stereoscopic/3D vision;<br>• each eye gives a slightly different image/view;<br>• allows better distance/speed judgement/wider field of view; | Allow better judgement of depth for 3D vision | 3 |

Total 7 marks

## Type MS · Explain stereoscopic vision and how it helps judge distance

### Answer 1 of 3 · Question 1 part b · 3 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain stereoscopic vision and how it helps judge distance

Answer 2 of 3 · Question 5 part b · 3 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 5 (a) | | *near* | *distant* | |
| | lens | thick | thin; | |
| | circular ciliary muscle | contracted | relaxed; | |
| | suspensory ligaments | loose | stretched; | 3 |
| (b) | • two eyes;<br>• each giving slightly different view;<br>• allows better judgement of distances / depth; | | | 3 |
| (c) | any six from<br>• test with two eyes then one;<br>• use same person/people of similar age/same eye condition;<br>• (throw and) catch a ball;<br>• 10 times/suitable repeats;<br>• record number of catches<br>• ref control variable e.g. lighting/wind condition/same size ball/throw from the same distance; | | | 6 |

Total 12 marks

## Type MS · Explain stereoscopic vision and how it helps judge distance

### Answer 2 of 3 · Question 5 part b · 3 marks

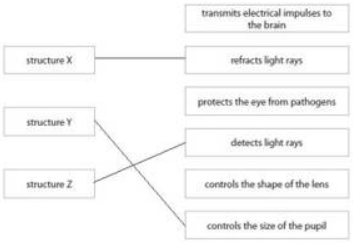
2022 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

# Type MS · Explain stereoscopic vision and how it helps judge distance

## Answer 3 of 3 · Question 1 part a.ii · 1 mark

2023 JAN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| number | | | | | |
|---|---|---|---|---|---|
| 1 | (a) | (i) |  | | 3 |
| | | (ii) | A;<br>B, C and D do not describe stereoscopic vision. | | 1 |
| | (b) | (i) | build-up of protein/cloudy/opaque;<br>in the lens; | | 2 |
| | | (ii) | surgery/operation to replace the lens/lens replacement; | | 1 |
| | | (iii) | 39 000000 x 43 ÷ 100;<br>16.77 million/16.8 million/16 770 000; | | 2 |
| | | (iv) | vitamin A; | | 1 |

Total 10 marks

## Type MS · Explain stereoscopic vision and how it helps judge distance

### Answer 3 of 3 · Question 1 part a.ii · 1 mark

2023 JAN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|